GATE 2014 Set 4 - Electronics and Communication Engineering

Answers are from the official answer key supplied with the paper. Worked solutions are given where available.

GATE 2014 Set 4 | Q.01

Q.1 Which of the following options is the closest in meaning to the word underlined in the sentence below?

In a democracy, everybody has the freedom to disagree with the government.

(A) dissent (B) descent (C) decent (D) decadent

Answer: A

GATE 2014 Set 4 | Q.02

Q.2 After the discussion, Tom said to me, 'Please revert!'. He expects me to _____.

(A) retract (B) get back to him (C) move in reverse (D) retreat

Answer: B

GATE 2014 Set 4 | Q.03

Q.3 While receiving the award, the scientist said, "I feel vindicated". Which of the following is closest in meaning to the word 'vindicated'?

(A) punished (B) substantiated (C) appreciated (D) chastened

Answer: B

GATE 2014 Set 4 | Q.04

Q.4 Let $f(x, y) = x^n y^m = P$. If x is doubled and y is halved, the new value of f is

(A) $2^{n-m}P$ (B) $2^{m-n}P$ (C) $2(n-m)P$ (D) $2(m-n)P$

Answer: A

GATE 2014 Set 4 | Q.05

Q.5 In a sequence of 12 consecutive odd numbers, the sum of the first 5 numbers is 425. What is the sum of the last 5 numbers in the sequence?

Answer: 495 to 495

GATE 2014 Set 4 | Q.06

Q.6 Find the next term in the sequence: 13M, 17Q, 19S, __

(A) 21W (B) 21V (C) 23W (D) 23V

Answer: C

GATE 2014 Set 4 | Q.07

Q.7 If 'KCLFTSB' stands for 'best of luck' and 'SHSWDG' stands for 'good wishes', which of the following indicates 'ace the exam'?

(A) MCHTX (B) MXHTC (C) XMHCT (D) XMHTC

Answer: B

GATE 2014 Set 4 | Q.08

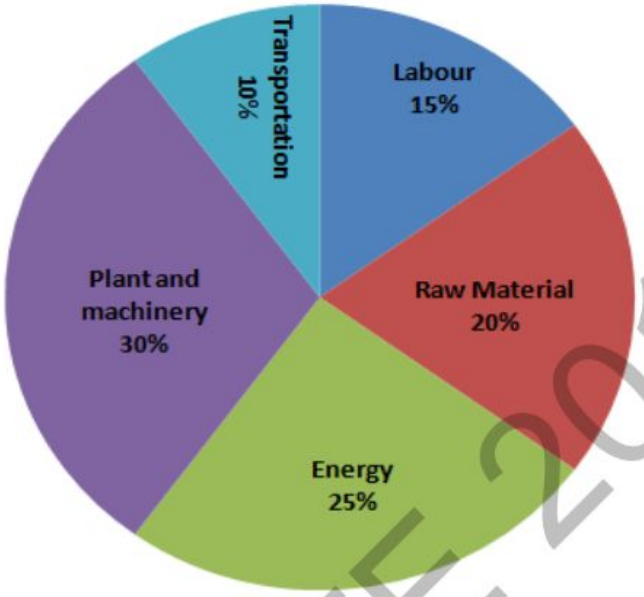
Q.8 Industrial consumption of power doubled from 2000-2001 to 2010-2011. Find the annual rate of increase in percent assuming it to be uniform over the years.

(A) 5.6 (B) 7.2 (C) 10.0 (D) 12.2

Answer: B

GATE 2014 Set 4 | Q.09

Q.9 A firm producing air purifiers sold 200 units in 2012. The following pie chart presents the share of raw material, labour, energy, plant & machinery, and transportation costs in the total manufacturing cost of the firm in 2012. The expenditure on labour in 2012 is Rs. 4,50,000. In 2013, the raw material expenses increased by 30% and all other expenses increased by 20%. What is the percentage increase in total cost for the company in 2013?



Answer: 22 to 22

GATE 2014 Set 4 | Q.10

Q.10 A five digit number is formed using the digits 1,3,5,7 and 9 without repeating any of them. What is the sum of all such possible five digit numbers?

(A) 6666660 (B) 6666600 (C) 6666666 (D) 6666606

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Answer: B

GATE 2014 Set 4 | Q.01

Q.1 The series $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges to

(A) 2 ln 2 (B) $\sqrt{2}$ (C) 2 (D) e

Answer: D

GATE 2014 Set 4 | Q.02

Q.2 The magnitude of the gradient for the function $f(x, y, z) = x^2 + 3y^2 + z^3$ at the point (1,1,1) is _____.

Answer: 6.8 to 7.2

GATE 2014 Set 4 | Q.03

Q.3 Let X be a zero mean unit variance Gaussian random variable. $E[|X|]$ is equal to _____

Answer: 0.79 to 0.81

GATE 2014 Set 4 | Q.04

Q.4 If a and b are constants, the most general solution of the differential equation

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + x = 0 \quad \text{is}$$

(A) ae^{-t} (B) $ae^{-t} + bte^{-t}$ (C) $ae^{t} + bte^{-t}$ (D) ae^{-2t}

Answer: B

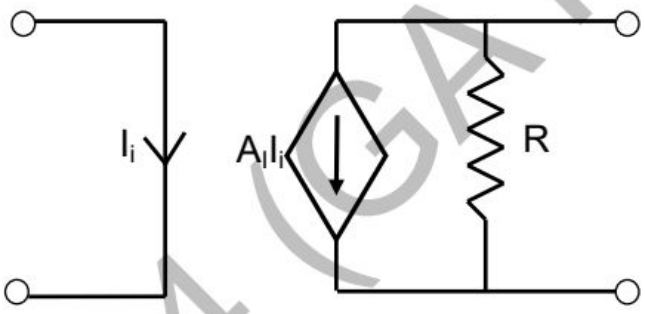
GATE 2014 Set 4 | Q.05

Q.5 The directional derivative of $f(x, y) = \frac{xy}{\sqrt{2}}(x + y)$ at (1, 1) in the direction of the unit vector at an angle of $\frac{\pi}{4}$ with y-axis, is given by _____ .

Answer: 2.99 to 3.01

GATE 2014 Set 4 | Q.06

Q.6 The circuit shown in the figure represents a

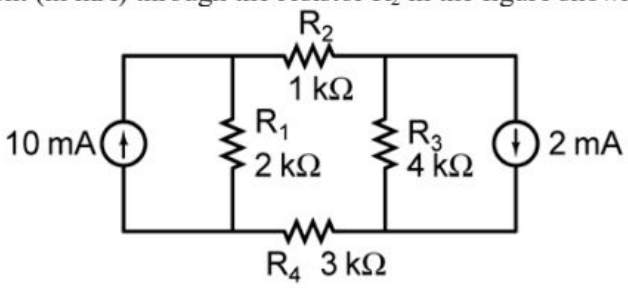


(A) voltage controlled voltage source
(B) voltage controlled current source
(C) current controlled current source
(D) current controlled voltage source

Answer: C

GATE 2014 Set 4 | Q.07

Q.7 The magnitude of current (in mA) through the resistor R_2 in the figure shown is_____.



Answer: 2.79 to 2.81

GATE 2014 Set 4 | Q.08

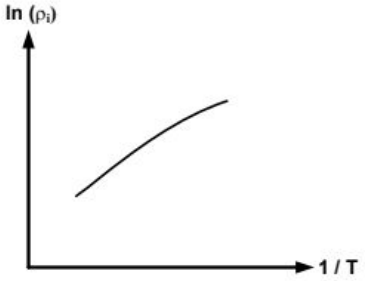
Q.8 At T = 300 K, the band gap and the intrinsic carrier concentration of GaAs are 1.42 eV and 10^6 cm^{-3}, respectively. In order to generate electron hole pairs in GaAs, which one of the wavelength (λ_C) ranges of incident radiation, is most suitable? (Given that: Plank's constant is 6.62×10^{-34} J-s, velocity of light is 3×10^{10} cm/s and charge of electron is 1.6×10^{-19} C)

(A) $0.42\ \mu m < \lambda_C < 0.87\ \mu m$
(B) $0.87\ \mu m < \lambda_C < 1.42\ \mu m$
(C) $1.42\ \mu m < \lambda_C < 1.62\ \mu m$
(D) $1.62\ \mu m < \lambda_C < 6.62\ \mu m$

Answer: A

GATE 2014 Set 4 | Q.09

Q.9 In the figure, ln (ρ_i) is plotted as a function of 1/T, where ρ_i is the intrinsic resistivity of silicon, T is the temperature, and the plot is almost linear.



The slope of the line can be used to estimate

(A) band gap energy of silicon (E_g)
(B) sum of electron and hole mobility in silicon ($\mu_n + \mu_p$)
(C) reciprocal of the sum of electron and hole mobility in silicon $(\mu_n + \mu_p)^{-1}$
(D) intrinsic carrier concentration of silicon (n_i)

Answer: A

GATE 2014 Set 4 | Q.10

Q.10 The cut-off wavelength (in μm) of light that can be used for intrinsic excitation of a semiconductor material of bandgap E_g= 1.1 eV is _____

Answer: 1.12 to 1.14

GATE 2014 Set 4 | Q.11

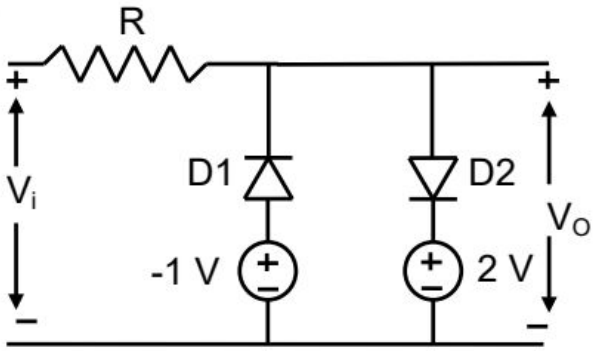
Q.11 If the emitter resistance in a common-emitter voltage amplifier is not bypassed, it will

(A) reduce both the voltage gain and the input impedance
(B) reduce the voltage gain and increase the input impedance
(C) increase the voltage gain and reduce the input impedance
(D) increase both the voltage gain and the input impedance

Answer: B

GATE 2014 Set 4 | Q.12

Q.12 Two silicon diodes, with a forward voltage drop of 0.7 V, are used in the circuit shown in the figure. The range of input voltage V_i for which the output voltage $V_o = V_i$, is

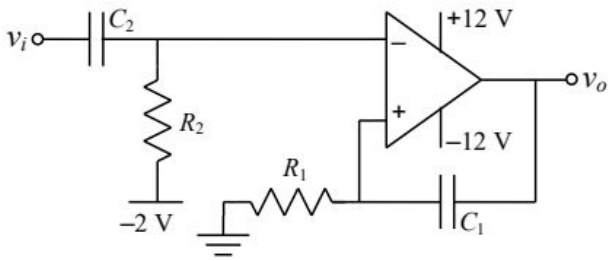


(A) $-0.3\,V < V_i < 1.3\,V$
(B) $-0.3\,V < V_i < 2\,V$
(C) $-1.0\,V < V_i < 2.0\,V$
(D) $-1.7\,V < V_i < 2.7\,V$

Answer: D

GATE 2014 Set 4 | Q.13

Q.13 The circuit shown represents



(A) a bandpass filter
(B) a voltage controlled oscillator
(C) an amplitude modulator
(D) a monostable multivibrator

Answer: D

GATE 2014 Set 4 | Q.14

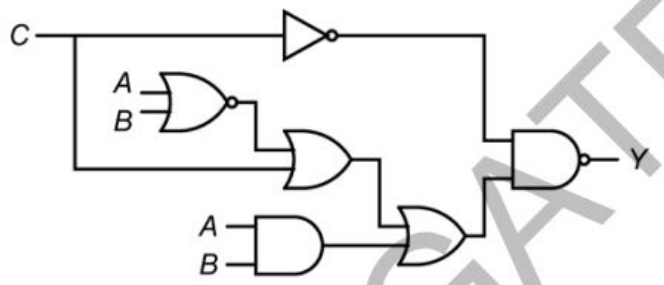
Q.14 For a given sample-and-hold circuit, if the value of the hold capacitor is increased, then

(A) droop rate decreases and acquisition time decreases
(B) droop rate decreases and acquisition time increases
(C) droop rate increases and acquisition time decreases
(D) droop rate increases and acquisition time increases

Answer: B

GATE 2014 Set 4 | Q.15

Q.15 In the circuit shown in the figure, if $C = 0$, the expression for Y is



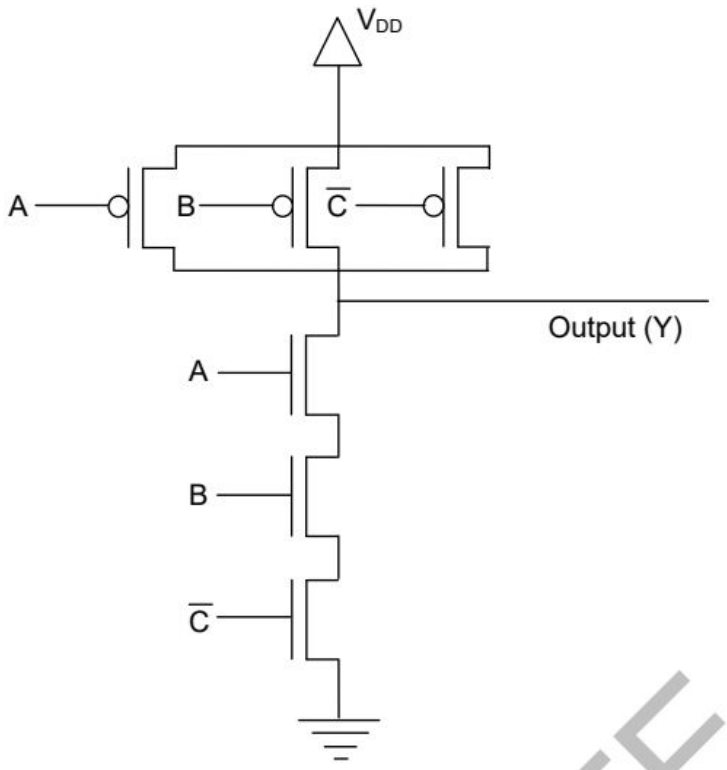
(A) $Y = A\,\bar{B} + \bar{A}\,B$ (B) $Y = A + B$ (C) $Y = \bar{A} + \bar{B}$ (D) $Y = A\,B$

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Answer: A

GATE 2014 Set 4 | Q.16

Q.16 The output (Y) of the circuit shown in the figure is



(A) $\overline{A}+\overline{B}+C$

(B) $A+\overline{B}\cdot\overline{C}+A\cdot\overline{C}$

(C) $\overline{A}+B+\overline{C}$

(D) $A\cdot B\cdot\overline{C}$

Answer: A

GATE 2014 Set 4 | Q.17

Q.17 A Fourier transform pair is given by

$$\left(\frac{2}{3}\right)^n u[n+3] \overset{FT}{\Leftrightarrow} \frac{A\,e^{-j6\pi f}}{1-\left(\frac{2}{3}\right)e^{-j2\pi f}}$$

where $u[n]$ denotes the unit step sequence. The values of A is _____.

Answer: 3.36 to 3.39

GATE 2014)

GATE 2014 Set 4 | Q.18

Q.18 A real-valued signal $x(t)$ limited to the frequency band $|f| \leq \frac{W}{2}$ is passed through a linear time invariant system whose frequency response is

$$H(f) = \begin{cases} e^{-j4\pi f}, & |f| \leq \frac{W}{2} \\ 0, & |f| > \frac{W}{2} \end{cases}.$$

The output of the system is

(A) $x(t+4)$ (B) $x(t-4)$ (C) $x(t+2)$ (D) $x(t-2)$

Answer: D

GATE 2014 Set 4 | Q.19

Q.19 The sequence $x[n] = 0.5^n\, u[n]$, where $u[n]$ is the unit step sequence, is convolved with itself to obtain $y[n]$. Then $\sum_{n=-\infty}^{+\infty} y[n]$ is _____.

Answer: 3.9 to 4.1

GATE 2014 Set 4 | Q.20

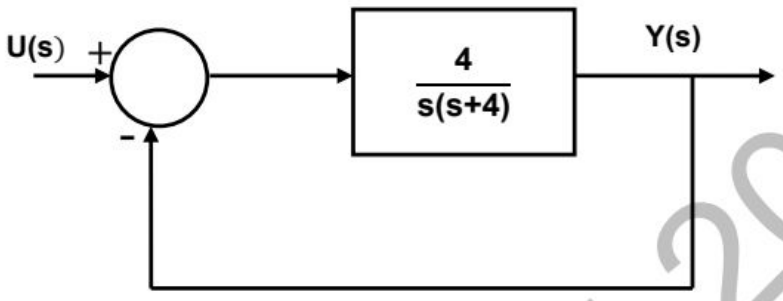
Q.20 In a Bode magnitude plot, which one of the following slopes would be exhibited at high frequencies by a 4th order all-pole system?

(A) – 80 dB/decade (B) – 40 dB/decade (C) +40 dB/decade (D) +80 dB/decade

Answer: A

GATE 2014 Set 4 | Q.21

Q.21 For the second order closed-loop system shown in the figure, the natural frequency (in rad/s) is



(A) 16 (B) 4 (C) 2 (D) 1

Answer: C

GATE 2014 Set 4 | Q.22

Q.22 If calls arrive at a telephone exchange such that the time of arrival of any call is independent of the time of arrival of earlier or future calls, the probability distribution function of the total number of calls in a fixed time interval will be

(A) Poisson (B) Gaussian (C) Exponential (D) Gamma

Answer: A

GATE 2014 Set 4 | Q.23

Q.23 In a double side-band (DSB) full carrier AM transmission system, if the modulation index is doubled, then the ratio of total sideband power to the carrier power increases by a factor of _____.

Answer: 3.95 to 4.05

GATE 2014 Set 4 | Q.24

Q.24 For an antenna radiating in free space, the electric field at a distance of 1 km is found to be 12 mV/m. Given that intrinsic impedance of the free space is $120\pi\ \Omega$, the magnitude of average power density due to this antenna at a distance of 2 km from the antenna (in nW/m^2) is ____.

Answer: 47.6 to 47.8

GATE 2014 Set 4 | Q.25

Q.25 Match column A with column B.

Column A	**Column B**
1. Point electromagnetic source	P. Highly directional
2. Dish antenna	Q. End fire
3. Yagi-Uda antenna	R. Isotropic

(A) $1 \rightarrow P$, $2 \rightarrow Q$, $3 \rightarrow R$

(B) $1 \rightarrow R$, $2 \rightarrow P$, $3 \rightarrow Q$

(C) $1 \rightarrow Q$, $2 \rightarrow P$, $3 \rightarrow R$

(D) $1 \rightarrow R$, $2 \rightarrow Q$, $3 \rightarrow P$

Answer: B

GATE 2014 Set 4 | Q.26

Q.26 With initial values $y(0) = y'(0) = 1$, the solution of the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$$

at $x = 1$ is _____.

Answer: 0.53 to 0.55

GATE 2014 Set 4 | Q.27

Q.27 Parcels from sender S to receiver R pass sequentially through two post-offices. Each post-office has a probability $\frac{1}{5}$ of losing an incoming parcel, independently of all other parcels. Given that a parcel is lost, the probability that it was lost by the second post-office is_____.

Answer: 0.43 to 0.45

GATE 2014 Set 4 | Q.28

Q.28 The unilateral Laplace transform of $f(t)$ is $\frac{1}{s^2+s+1}$. Which one of the following is the unilateral Laplace transform of $g(t) = t \cdot f(t)$?

(A) $\frac{-s}{(s^2+s+1)^2}$ (B) $\frac{-(2s+1)}{(s^2+s+1)^2}$

(C) $\frac{s}{(s^2+s+1)^2}$ (D) $\frac{2s+1}{(s^2+s+1)^2}$

Answer: D

GATE 2014 Set 4 | Q.29

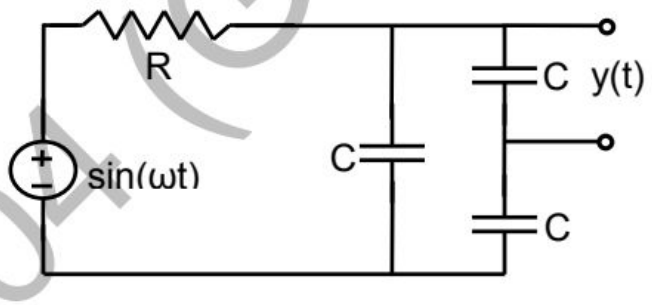
Q.29 For a right angled triangle, if the sum of the lengths of the hypotenuse and a side is kept constant, in order to have maximum area of the triangle, the angle between the hypotenuse and the side is

(A) 12° (B) 36° (C) 60° (D) 45°

Answer: C

GATE 2014 Set 4 | Q.30

Q.30 The steady state output of the circuit shown in the figure is given by $y(t) = A(\omega)\sin(\omega t + \phi(\omega))$. If the amplitude $|A(\omega)| = 0.25$, then the frequency ω is

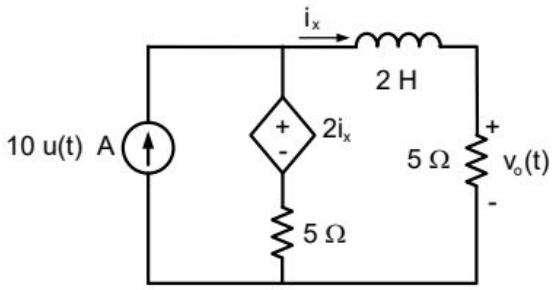


(A) $\frac{1}{\sqrt{3}\,R\,C}$ (B) $\frac{2}{\sqrt{3}\,R\,C}$ (C) $\frac{1}{R\,C}$ (D) $\frac{2}{R\,C}$

Answer: B

GATE 2014 Set 4 | Q.31

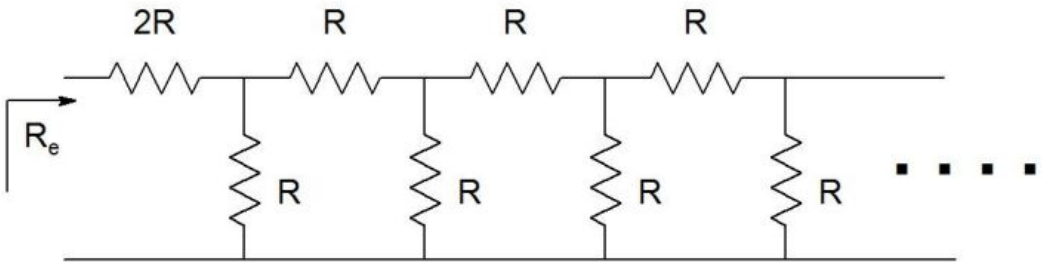
Q.31 In the circuit shown in the figure, the value of $v_0(t)$ (in Volts) for $t \to \infty$ is _____.



Answer: 31.24 to 31.26

GATE 2014 Set 4 | Q.32

Q.32 The equivalent resistance in the infinite ladder network shown in the figure, is R_e.

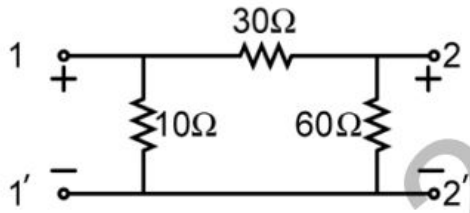


The value of R_e/R is _____

Answer: 2.60 to 2.64

GATE 2014 Set 4 | Q.33

Q.33 For the two-port network shown in the figure, the impedance (Z) matrix (in Ω) is



(A) $\begin{bmatrix} 6 & 24 \\ 42 & 9 \end{bmatrix}$ (B) $\begin{bmatrix} 9 & 8 \\ 8 & 24 \end{bmatrix}$ (C) $\begin{bmatrix} 9 & 6 \\ 6 & 24 \end{bmatrix}$ (D) $\begin{bmatrix} 42 & 6 \\ 6 & 60 \end{bmatrix}$

Answer: C

GATE 2014 Set 4 | Q.34

Q.34 Consider a silicon sample doped with $N_D = 1\times10^{15}/cm^3$ donor atoms. Assume that the intrinsic carrier concentration $n_i = 1.5\times10^{10}/cm^3$. If the sample is additionally doped with $N_A = 1\times10^{18}/cm^3$ acceptor atoms, the approximate number of electrons/cm^3 in the sample, at T=300 K, will be _____.

Answer: 224.9 to 225.1

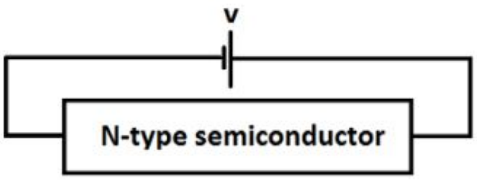
GATE 2014 Set 4 | Q.35

Q.35 Consider two BJTs biased at the same collector current with area $A_1 = 0.2\ \mu m \times 0.2\ \mu m$ and $A_2 = 300\ \mu m \times 300\ \mu m$. Assuming that all other device parameters are identical, $kT/q = 26$ mV, the intrinsic carrier concentration is $1 \times 10^{10}\ cm^{-3}$, and $q = 1.6 \times 10^{-19}$ C, the difference between the base-emitter voltages (in mV) of the two BJTs (i.e., $V_{BE1} - V_{BE2}$) is _____.

Answer: 378 to 381

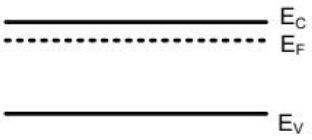
GATE 2014 Set 4 | Q.36

Q.36 An N-type semiconductor having uniform doping is biased as shown in the figure.



If E_C is the lowest energy level of the conduction band, E_V is the highest energy level of the valance band and E_F is the Fermi level, which one of the following represents the energy band diagram for the biased N-type semiconductor?

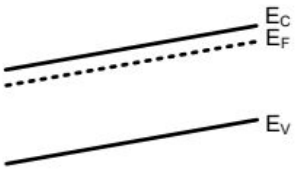
(A)



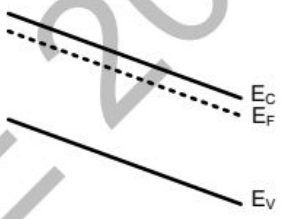
(B)



(C)



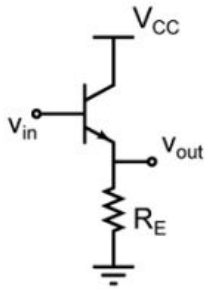
(D)



Answer: D

GATE 2014 Set 4 | Q.37

Q.37 Consider the common-collector amplifier in the figure (bias circuitry ensures that the transistor operates in forward active region, but has been omitted for simplicity). Let I_C be the collector current, V_{BE} be the base-emitter voltage and V_T be the thermal voltage. Also, g_m and r_o are the small-signal transconductance and output resistance of the transistor, respectively. Which one of the following conditions ensures a nearly constant small signal voltage gain for a wide range of values of R_E?



(A) $g_m R_E \ll 1$ (B) $I_C R_E \gg V_T$ (C) $g_m r_o \gg 1$ (D) $V_{BE} \gg V_T$

Answer: B

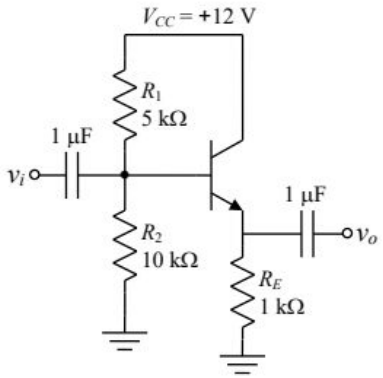
GATE 2014 Set 4 | Q.38

Q.38 A BJT in a common-base configuration is used to amplify a signal received by a 50 Ω antenna. Assume kT/q = 25 mV. The value of the collector bias current (in mA) required to match the input impedance of the amplifier to the impedance of the antenna is_____.

Answer: 0.49 to 0.51

GATE 2014 Set 4 | Q.39

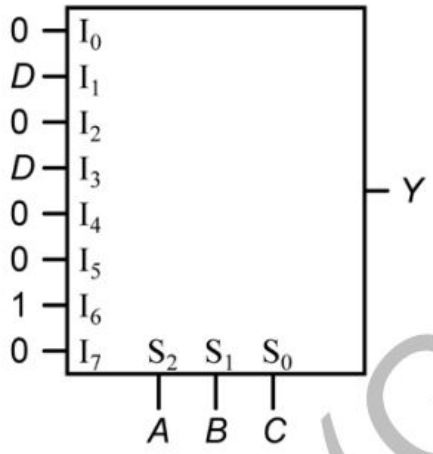
Q.39 For the common collector amplifier shown in the figure, the BJT has high β, negligible $V_{CE(sat)}$, and $V_{BE} = 0.7$ V. The maximum undistorted peak-to-peak output voltage v_o (in Volts) is _____.



Answer: 9.39 to 9.41

GATE 2014 Set 4 | Q.40

Q.40 An 8-to-1 multiplexer is used to implement a logical function Y as shown in the figure. The output Y is given by

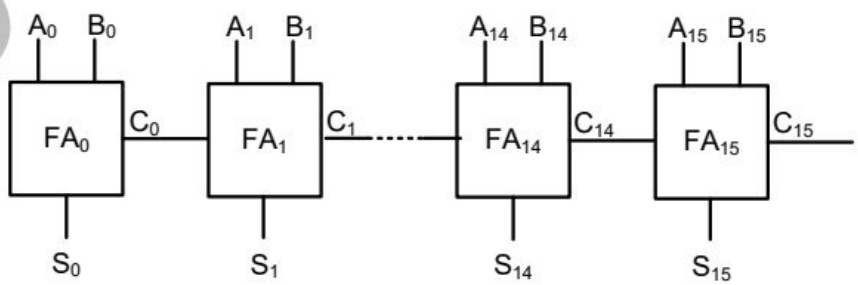


(A) $Y = A\bar{B}C + A\bar{C}D$
(B) $Y = \bar{A}BC + A\bar{B}D$
(C) $Y = AB\bar{C} + \bar{A}CD$
(D) $Y = \bar{A}\bar{B}D + A\bar{B}C$

Answer: C

GATE 2014 Set 4 | Q.41

Q.41 A 16-bit ripple carry adder is realized using 16 identical full adders (FA) as shown in the figure. The carry-propagation delay of each FA is 12 ns and the sum-propagation delay of each FA is 15 ns. The worst case delay (in ns) of this 16-bit adder will be _____.



Answer: 194.9 to 195.1

GATE 2014 Set 4 | Q.42

Q.42 An 8085 microprocessor executes "STA 1234H" with starting address location 1FFEH (STA copies the contents of the Accumulator to the 16-bit address location). While the instruction is fetched and executed, the sequence of values written at the address pins $A_{15} - A_8$ is

(A) 1FH, 1FH, 20H, 12H
(B) 1FH, FEH, 1FH, FFH, 12H
(C) 1FH, 1FH, 12H, 12H
(D) 1FH, 1FH, 12H, 20H, 12H

Answer: A

GATE 2014 Set 4 | Q.43

Q.43 A stable linear time invariant (LTI) system has a transfer function $H(s) = \frac{1}{s^2+s-6}$. To make this system causal it needs to be cascaded with another LTI system having a transfer function $H_1(s)$. A correct choice for $H_1(s)$ among the following options is

(A) $s+3$ (B) $s-2$ (C) $s-6$ (D) $s+1$

Answer: B

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GATE 2014 Set 4 | Q.44

Q.44 A causal LTI system has zero initial conditions and impulse response $h(t)$. Its input $x(t)$ and output $y(t)$ are related through the linear constant-coefficient differential equation

$$\frac{d^2y(t)}{dt^2} + \alpha\frac{dy(t)}{dt} + \alpha^2 y(t) = x(t).$$

Let another signal $g(t)$ be defined as

$$g(t) = \alpha^2 \int_0^t h(\tau)d\tau + \frac{dh(t)}{dt} + \alpha h(t).$$

If $G(s)$ is the Laplace transform of $g(t)$, then the number of poles of $G(s)$ is _____.

Answer: 0.99 to 1.01

GATE 2014 Set 4 | Q.45

Q.45 The N-point DFT X of a sequence $x[n]$, $0 \le n \le N-1$ is given by

$$X[k] = \frac{1}{\sqrt{N}}\sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}nk}, \qquad 0 \le k \le N-1.$$

Denote this relation as $X = DFT(x)$. For $N = 4$, which one of the following sequences satisfies $DFT(DFT(x)) = x$?

(A) $x = [1\ \ 2\ \ 3\ \ 4]$
(B) $x = [1\ \ 2\ \ 3\ \ 2]$
(C) $x = [1\ \ 3\ \ 2\ \ 2]$
(D) $x = [1\ \ 2\ \ 2\ \ 3]$

Answer: B

GATE 2014 Set 4 | Q.46

Q.46 The state transition matrix $\boldsymbol{\phi}$(t) of a system $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$ is

(A) $\begin{bmatrix}t & 1\\ 1 & 0\end{bmatrix}$ (B) $\begin{bmatrix}1 & 0\\ t & 1\end{bmatrix}$ (C) $\begin{bmatrix}0 & 1\\ 1 & t\end{bmatrix}$ (D) $\begin{bmatrix}1 & t\\ 0 & 1\end{bmatrix}$

Answer: D

GATE 2014 Set 4 | Q.47

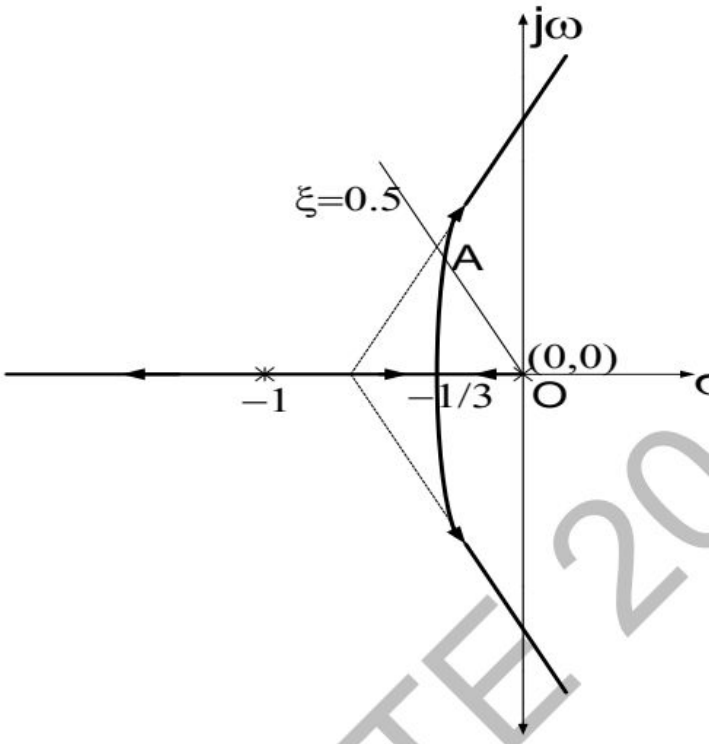
Q.47 Consider a transfer function $G_p(s) = \frac{ps^2+3ps-2}{s^2+(3+p)s+(2-p)}$ with p a positive real parameter. The maximum value of p until which G_p remains stable is _____.

Answer: 1.9 to 2.1

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GATE 2014 Set 4 | Q.48

Q.48 The characteristic equation of a unity negative feedback system is $1 + KG(s) = 0$. The open loop transfer function $G(s)$ has one pole at 0 and two poles at -1. The root locus of the system for varying K is shown in the figure.



The constant damping ratio line, for ξ=0.5, intersects the root locus at point A. The distance from the origin to point A is given as 0.5. The value of K at point A is _____ .

Answer: 0.32 to 0.41

GATE 2014 Set 4 | Q.49

Q.49 Consider a communication scheme where the binary valued signal X satisfies $P\{X = +1\} = 0.75$ and $P\{X = -1\} = 0.25$. The received signal $Y = X + Z$, where Z is a Gaussian random variable with zero mean and variance σ^2. The received signal Y is fed to the threshold detector. The output of the threshold detector $\hat{X}$ is:

$$\hat{X} = \begin{cases} +1, & Y > \tau \\ -1, & Y \leq \tau. \end{cases}$$

To achieve a minimum probability of error $P\{\hat{X} \neq X\}$, the threshold τ should be

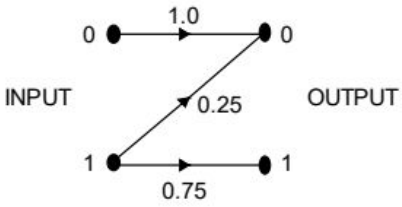
(A) strictly positive
(B) zero
(C) strictly negative
(D) strictly positive, zero, or strictly negative depending on the nonzero value of σ^2

Answer: C

EC04 (GATE 2014)

GATE 2014 Set 4 | Q.50

Q.50 Consider the Z-channel given in the figure. The input is 0 or 1 with equal probability.



If the output is 0, the probability that the input is also 0 equals _____

Answer: 0.79 to 0.81

GATE 2014 Set 4 | Q.51

Q.51 An M-level PSK modulation scheme is used to transmit independent binary digits over a band-pass channel with bandwidth 100 kHz. The bit rate is 200 kbps and the system characteristic is a raised-cosine spectrum with 100% excess bandwidth. The minimum value of M is _____.

Answer: 15.9 to 16.1

GATE 2014 Set 4 | Q.52

Q.52 Consider a discrete-time channel $Y = X + Z$, where the additive noise Z is signal-dependent. In particular, given the transmitted symbol $X \in \{-a, +a\}$ at any instant, the noise sample Z is chosen independently from a Gaussian distribution with mean βX and unit variance. Assume a threshold detector with zero threshold at the receiver.

When $\beta = 0$, the BER was found to be $Q(a) = 1 \times 10^{-8}$.
$\left(Q(v) = \frac{1}{\sqrt{2\pi}} \int_v^{\infty} e^{-u^2/2}\, du$, and for $v > 1$, use $Q(v) \approx e^{-v^2/2}\right)$

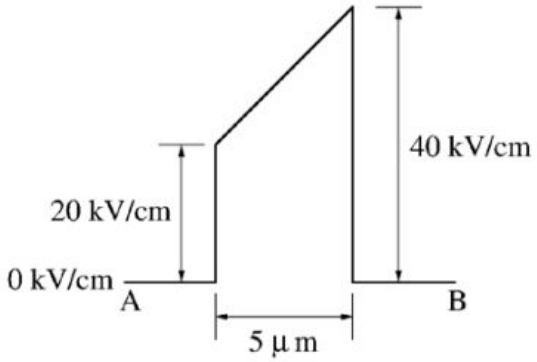
When $\beta = -0.3$, the BER is closest to

(A) 10^{-7} (B) 10^{-6} (C) 10^{-4} (D) 10^{-2}

Answer: C

GATE 2014 Set 4 | Q.53

Q.53 The electric field (assumed to be one-dimensional) between two points A and B is shown. Let ψ_A and ψ_B be the electrostatic potentials at A and B, respectively. The value of $\psi_B - \psi_A$ in Volts is _____.



Answer: -15.1 to -14.9

GATE 2014 Set 4 | Q.54

Q.54 Given $\vec{F} = z\hat{a}_x + x\hat{a}_y + y\hat{a}_z$. If S represents the portion of the sphere $x^2 + y^2 + z^2 = 1$ for $z \geq 0$, then $\int_s \nabla \times \vec{F} \cdot \vec{ds}$ is_____.

Answer: 3.13 to 3.15

Q.55 If $\vec{E} = -(2y^3 - 3yz^2)\hat{x} - (6xy^2 - 3xz^2)\hat{y} + (6xyz)\hat{z}$ is the electric field in a source free region, a valid expression for the electrostatic potential is

(A) $xy^3 - yz^2$ (B) $2xy^3 - xyz^2$ (C) $y^3 + xyz^2$ (D) $2xy^3 - 3xyz^2$

END OF THE QUESTION PAPER

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Answer: D